

Certified Identifiability of Fractional ECG Harmonic Surrogates from Multi-Rate Records

Anonymous authors

Abstract

We study when a scalar fractional harmonic surrogate can be identified from surface-ECG windows acquired at different heart rates. The object of inference is an effective input–output order, not an internal cardiac state or a physiological fractional mechanism. For the asymptotic harmonic branch of a dimensionless Caputo model with unknown common complex morphology and positive real damping, we prove that one exact complex rate response is insufficient and two are globally sufficient. The proof constructs a complete common-gain quotient whose physical image is explicit and whose inverse is unique on $(0, 1] \times \mathbb{R}_{>0}$. We extend the result to an exact multi-rate consistency test, and prove two limits: unrestricted rate-specific complex morphology destroys order identifiability, while any positive independent morphology disk prevents singleton recovery locally in the two-rate known-damping problem. For finite records, we connect fixed-phase complex weighted least squares, independently calibrated residual bounds, joint response-disk set inversion, validated Gram margins, and a fail-closed replay certificate. Specializing the standard reliable-set minimax principle to the gauge-reduced response disks gives an exact separation threshold for rate-window design. Deterministic sweeps recovered all 1,260 exact two-rate cases, preserved all common-gain invariants, and detected all 1,260 independently valid but three-rate-inconsistent controls. A design-locked four-record Fantasia engineering pilot replayed 1,859/1,859 WLS records, but all four fractional fits terminated at $\alpha = 1$ and none materially improved on the nested $\alpha = 1$ model. The distinct fractional-order hypothesis is therefore **NOT_SUPPORTED**; because several acquisition and model-error primitives remain uncalibrated, all four real records are also **NOT_CERTIFIABLE**. The negative result is part of the proposed certification discipline rather than evidence for a cardiac digital twin.

Keywords: fractional-order systems; structural identifiability; set-membership estimation; electrocardiography; validated numerics; experiment design.

1 Introduction

Fractional differential models provide compact representations of memory and power-law relaxation. They have been used in ECG signal generators and cardiac rhythm models [13, 14, 15], while Kalman-type filters for fractional systems already estimate states, parameters, and order [3, 4, 5]. These precedents make it scientifically inappropriate to present either a fractional ECG model or a fractional extended Kalman filter as the contribution by itself. The harder control question is prior to filtering: under which phase, morphology, damping, and uncertainty contracts is the effective order identifiable from a finite collection of ECG rate windows?

The distinction matters because a surface ECG is a low-dimensional projection of a distributed electrophysiological process. Even detailed cardiac-twin inverse problems can have severe nonuniqueness [16, 17]. A flexible rate-specific gain can likewise make a simple harmonic surrogate fit every candidate order. A point estimate returned by an optimizer therefore does not establish either identifiability or physiological meaning. A defensible workflow must expose the nuisance gauge, give a finite-record coverage statement, and refuse a conclusion when the required error bounds are absent.

This paper develops such a workflow for the declared scalar harmonic model. Its contributions are deliberately narrow.

1. We construct a complete common-complex-gain quotient and prove global two-rate recovery of order and damping. One rate is strictly insufficient, and a third rate supplies a redundant exact consistency equation.
2. We characterize the boundary of the claim: rate-specific complex morphology eliminates order information; positive independent morphology disks preclude local point recovery; and finite frequency responses cannot distinguish a fractional mechanism from a stable rational interpolant.
3. We formulate the finite-data problem as a joint response-disk inversion. A conditional data-to-disk theorem separates WLS geometry from independently calibrated history, dwell, phase, sampling, front-end, noise, and model-discrepancy budgets. Validated Gram and replay checks fail closed rather than silently shrinking uncertainty.
4. We specialize the standard reliable-set minimax identity and exact product-disk separation threshold to the common-gauge ECG geometry and its certified R2 radii, turning heart-rate/window selection into an operational worst-case set-diameter design problem.
5. We report both positive synthetic checks and a negative four-record Fantasia pilot. The latter supports the reproducibility of the signal processing path, but not a distinct fractional order and not a certified real-ECG response disk.

Detailed proofs, validated-continuation constructions, and all component bounds are given in the accompanying theory-core supplement. The present paper retains only the results needed to understand the control contribution and its empirical boundary.

For cross-document navigation, the supplement numbers the quotient geometry, global two-rate inverse, and reference-pair consistency as S4, S5, and S7; the forward-disk and morphology-drift results as S13 and S15; the analytic selector tube and conditional WLS enclosure as S32 and S39; the exact-payload enclosure and R2 component/transfer results as S44, S46, and S47; and the reliable-set/design results as S52 and S54.

2 Related work and claim boundary

2.1 Fractional identification and set membership

Frequency-domain identifiability of fractional systems and its sensitivity limits are established topics [6]. Malti et al. studied set-membership estimation for an isomorphic first-order fractional transfer model using bounded complex frequency data [9]; Khemani et al. developed robust frequency-domain estimation under rectangular, polar, and disk uncertainty descriptions [10]. Robust fractional experiment design and general nonlinear set-membership design also predate this work [11, 12]. Consequently, neither complex disks, interval inversion, nor experiment design is claimed as a generic novelty here.

The specialization pursued below is the interaction among (i) a common unknown complex ECG morphology/lead gauge, (ii) positive real damping, (iii) multi-rate phase-locked harmonic responses, and (iv) a finite-record certificate whose evidence objects can be replayed. Validated linear algebra and parameter-dependent enclosures follow standard principles [20, 21, 22, 23]; our contribution is the model geometry and the composition semantics, not a new interval-Newton method.

2.2 Fractional ECG and Kalman models

The integer-order McSharry generator [1] and its nonlinear Bayesian filtering descendants [2] established dynamical ECG state-space modelling well before the present study. Fractional ECG generators and McSharry extensions have also been reported [13, 15]. Fractional Kalman filters have a substantial history [3, 4, 5]. We therefore do not claim a new ECG generator, a generic

fractional filter, observer stability under fractional-order mismatch, or direct recovery of cardiac tissue states.

Our model is a *surface-ECG harmonic surrogate*. Its order is an effective parameter inside a declared family. The present evidence does not meet the subject-specific synchronization and held-out predictive standard needed to call the construction a cardiac digital twin.

3 Problem formulation

3.1 Dimensionless harmonic surrogate

Fix a reference time $\tau_\star > 0$, set $\xi = t/\tau_\star$, and consider

$${}^C D_\xi^\alpha z(\xi) = -\lambda z(\xi) + \sum_{m \in \mathbb{Z}} q_m e^{im\theta(\xi)}. \quad (1)$$

Here λ is dimensionless and q_m has the measurement unit of z . We do not interpret (1) with a finite lower terminal as possessing an exact periodic orbit: time-invariant Caputo systems generally do not [7]. The response below is the infinite-history (Weyl) harmonic branch, equivalently the asymptotic periodic component of the finite-terminal Caputo solution [8]. Initial-condition, prehistory, and finite-dwell transients are separate terms in the data-to-response-disk coverage gate (R2). At one positive harmonic and rate r , this branch is

$$Z_r = \frac{q}{\lambda + (i\nu_r)^\alpha}, \quad (i\nu)^\alpha = \nu^\alpha e^{i\pi\alpha/2}, \quad (2)$$

with

$$\alpha \in I = [\alpha_-, 1], \quad 0 < \alpha_- < 1, \quad \lambda \in \Lambda \subseteq \mathbb{R}_{>0}, \quad q \in \mathbb{C} \setminus \{0\}.$$

The dimensionless frequency is $\nu_{r,m} = m\tau_\star\Omega_r$, where Ω_r is the base angular heart rate in rad/s. A common nonzero complex lead gain is absorbed into q . For real ECG, the negative harmonic is the conjugate of the positive one and does not add an independent model equation.

Assumption 3.1 (Declared information boundary). Rates are known, positive, and distinct. Responses share one R-peak phase anchor, one causal preprocessing chain, the principal fractional branch, and common (α, λ, q) . No uncalibrated rate-specific complex gain, phase, or delay is allowed. Any restitution or acquisition bound used in a certificate is calibrated independently of the identification residual.

The estimator may use a two-sided basis $\mathcal{K}_{\text{est}} \subset \mathbb{Z}$, including DC and conjugate negative harmonics. The identification targets $\mathcal{M} \subset \mathbb{N}_{>0}$ contain only positive harmonics. This distinction prevents DC or a duplicated conjugate coefficient from entering formulae that use $\log \nu$.

3.2 Finite-window estimator and response disks

In a digital window with exact integer endpoints $j_a < j_b$ spanning N complete detected RR intervals, let $j_n \in [j_a, j_b)$ be the selected sample indices, $w_n > 0$, and $\sum_n w_n = 1$. Freeze the whole-window phase

$$\hat{\phi}_n = 2\pi N \frac{j_n - j_a}{j_b - j_a}, \quad \Psi_{n,k} = e^{ik\hat{\phi}_n}, \quad k \in \mathcal{K}_{\text{est}}.$$

The implementation never reconstructs this phase through rounded binary64 seconds; timestamp uncertainty is budgeted separately. With $W = \text{diag}(w_n)$ and $G = \Psi^* W \Psi$, the joint complex WLS coefficient and front-end-corrected target are

$$\tilde{C} = G^{-1} \Psi^* W \mathbf{y}, \quad \tilde{Z}_m = \tilde{C}_m / \hat{H}_m. \quad (3)$$

The real-signal convention is $y = Z_0 + 2 \operatorname{Re} \sum_{m>0} Z_m e^{im\phi}$; a one-sided sinusoidal amplitude is not substituted for the complex Fourier coefficient. In finite-window formulae, $\hat{H}_m$ abbreviates the calibrated $\hat{H}_{r,m}$ for the current rate/window. Values may vary with physical frequency under the one frozen causal front end; no uncalibrated rate-specific gain is absorbed into the morphology.

The inference layer receives disks

$$|Z_{r,m} - \tilde{Z}_{r,m}| \leq \epsilon_{r,m}. \quad (4)$$

The centre and radius have different evidentiary roles: the centre is replayed from the named window payload, whereas every contribution to the radius needs an independent deterministic or simultaneous probabilistic contract.

3.3 Decision semantics

We distinguish four protocol-level outcomes:

`PASS_DETERMINISTIC`, `PASS_PROBABILISTIC`, `NOT_CERTIFIABLE`, `EXCLUDE_PROTOCOL`.

`PASS_PROBABILISTIC` requires a predeclared simultaneous-coverage contract and an explicit joint failure-probability bound. `EXCLUDE_PROTOCOL` means that the acquisition or preprocessing design violates the information boundary—for example, a noncausal target-window operation or an uncalibrated rate-specific amplitude/phase normalization—so the protocol is ineligible for inference. Missing bounds, singular Gram geometry, a wide nonempty order set, or a broken hash chain instead produce `NOT_CERTIFIABLE`; none is evidence that the model is false. The separate downstream relation `REJECT_MODEL` requires a complete joint outer proof of emptiness with no unresolved branch; it is not an R2 protocol status.

The empirical label `NOT_SUPPORTED`, used only in the held-out model comparison below, is not a fifth certificate relation. It means that the predeclared material-improvement threshold was not met. It is neither `NOT_CERTIFIABLE` nor model rejection, does not prove that $\alpha < 1$ or the full model family is false, and may coexist with `NOT_CERTIFIABLE`.

4 Structural identifiability and its limits

4.1 A complete common-gain quotient

For two nonzero responses let $W_r = Z_r^{-1}$, $D = W_2 - W_1$, and $C = W_1/D$ when $D \neq 0$. The number C is invariant under $(Z_1, Z_2) \mapsto (gZ_1, gZ_2)$ for any $g \in \mathbb{C} \setminus \{0\}$. It is also complete: equality of C for two ordered pairs implies that the pairs differ by one common nonzero complex gain.

Proposition 4.1 (One rate is insufficient). *For one exact response Z_1 , the order is not structurally identifiable.*

Proof. For every candidate $(\beta, \ell) \in I \times \Lambda$, the nonzero choice $q_{\beta, \ell} = Z_1[\ell + (i\nu_1)^\beta]$ reproduces Z_1 . $\square$

Theorem 4.2 (Global two-rate recovery modulo a common complex gauge). *Let $0 < \nu_1 < \nu_2$ and suppose two exact responses satisfy (2). Put $\varrho = \nu_2/\nu_1 > 1$. Then $D \neq 0$, $\operatorname{Im} C < 0$, and the true order is the unique $\beta \in (0, 1]$ satisfying*

$$(\varrho^\beta - 1) \left[\operatorname{Re} C + \operatorname{Im} C \cot \left(\frac{\pi\beta}{2} \right) \right] = 1. \quad (5)$$

After recovering α ,

$$q = \frac{e^{i\pi\alpha/2}(\nu_2^\alpha - \nu_1^\alpha)}{D}, \quad \lambda = \nu_1^\alpha e^{i\pi\alpha/2} [C(\varrho^\alpha - 1) - 1] \in \mathbb{R}_{>0}. \quad (6)$$

Equivalently, the quotient map

$$\Phi_{\varrho}(\alpha, \kappa) = \frac{1 + \kappa e^{-i\pi\alpha/2}}{\varrho^\alpha - 1}, \quad \kappa = \lambda/\nu_1^\alpha, \quad (7)$$

is one-to-one from $(0, 1] \times \mathbb{R}_{>0}$ onto

$$\mathcal{Q}_{\varrho} = \{u + iv : u \geq (\varrho - 1)^{-1}, v < 0\}.$$

It is a diffeomorphism between the corresponding interiors.

Proof idea. Inverse-response subtraction eliminates damping and gives $D = e^{i\pi\alpha/2}(\nu_2^\alpha - \nu_1^\alpha)/q$. Hence

$$C = \frac{1 + \kappa e^{-i\pi\alpha/2}}{\varrho^\alpha - 1}, \quad \kappa > 0.$$

For a candidate β , requiring the reconstructed damping to be real is exactly (5). Substitution of the true quotient reduces the candidate equation to

$$\varrho^\beta - \varrho^\alpha + \kappa(\varrho^\beta - 1) \frac{\sin[\pi(\beta - \alpha)/2]}{\sin(\pi\beta/2)} = 0.$$

Both terms have the sign of $\beta - \alpha$, proving uniqueness. Positivity of the reconstructed damping follows from $\text{Im } C < 0$. Surjectivity onto $\mathcal{Q}_{\varrho}$ follows by the same sign argument at the interval endpoints. Finally,

$$\det \frac{\partial(\text{Re } C, \text{Im } C)}{\partial(\alpha, \kappa)} = \frac{\varrho^\alpha \log \varrho \sin(\pi\alpha/2)}{(\varrho^\alpha - 1)^3} + \frac{\kappa\pi}{2(\varrho^\alpha - 1)^2} > 0,$$

which supplies the interior diffeomorphism. $\square$

The endpoint $\alpha = 1$ is physical and must not be rejected merely because it lies on the image boundary. Conversely, structural uniqueness is not a conditioning guarantee: no uniform noise margin exists as $\alpha \downarrow 0$ or $\varrho \downarrow 1$.

4.2 Multi-rate consistency

Theorem 4.3 (Reference-pair consistency). *Let $0 < \nu_1 < \dots < \nu_R$, $R \geq 2$, and let all exact responses be nonzero. For $r \geq 2$ define*

$$C_{1r} = \frac{W_1}{W_r - W_1}, \quad (\alpha_r, \kappa_r) = \Phi_{\nu_r/\nu_1}^{-1}(C_{1r}).$$

Assume every denominator is nonzero and every quotient lies in its physical image. All responses arise from one common (α, λ, q) if and only if (α_r, κ_r) is independent of r . The common parameters belong to the declared compact model if and only if additionally $\alpha_\star \in I$ and $\kappa_\star \nu_1^{\alpha_\star} \in \Lambda$.

Proof idea. Necessity follows by applying Theorem 4.2 to every pair. For sufficiency, the common inverse quotient gives each difference $W_r - W_1$ generated by the same α and q ; adding the reference equation reconstructs every W_r . Compact-prior membership is a separate check, not a consequence of structural inversion. $\square$

Corollary 4.4 (Geometric three-rate cross-ratio). *Let $\nu_2 = \rho\nu_1$ and $\nu_3 = \rho^2\nu_1$ with $\rho > 1$, and define*

$$Q = \frac{W_3 - W_1}{W_2 - W_1}.$$

Every response triple in the zero-drift model satisfies the damping- and morphology-free identity

$$Q = 1 + \rho^\alpha \in \mathbb{R}, \quad \alpha = \frac{\log(Q - 1)}{\log \rho}. \quad (8)$$

Conversely, an admissible reference pair $(1, 2)$ with inverse (α_*, κ_*) extends to the third response if and only if $Q = 1 + \rho^{\alpha_*}$. Together with $\alpha_* \in I$ and $\kappa_* \nu_1^{\alpha_*} \in \Lambda$, this is an exact three-rate model-consistency test.

Proof. Under the model, $W_r - W_1 = [(i\nu_r)^\alpha - (i\nu_1)^\alpha]/q$. The common factors $q^{-1}(i\nu_1)^\alpha$ cancel, giving $Q = (\rho^{2\alpha} - 1)/(\rho^\alpha - 1) = 1 + \rho^\alpha$. Conversely, the reference pair fixes (α_*, λ, q) ; the cross-ratio identity is exactly the remaining equation $W_3 - W_1$. $\square$

Corollary 4.4 makes the role of the third geometric rate explicit: it is a genuine redundant equation that can falsify the declared zero-drift family, not another rate-specific nuisance parameter to be fitted.

For multiple positive harmonics, the order must be common, while $\kappa_m \nu_{1,m}^\alpha = \lambda$ must agree across harmonics. Pairwise centre inversion is only an exact-data diagnostic; under disks the shared variables must be retained in one joint set.

4.3 No-go results

Proposition 4.5 (Rate-specific morphology destroys order information). *If an arbitrary nonzero complex q_r is allowed at every rate, then every candidate $(\beta, \ell) \in I \times \mathbb{R}_{>0}$ fits any finite collection of nonzero responses through $q_r = Z_r[\ell + (i\nu_r)^\beta]$.*

Thus per-window complex normalization is not innocuous preprocessing: it removes precisely the common-gauge relation that carries order information.

Theorem 4.6 (Positive independent morphology drift prevents point order). *Fix two rates and known $\lambda_0 > 0$. Let exact zero-drift responses be generated at $(\alpha_0, \lambda_0, q_0)$, and let the declared morphology domain be the compact annulus $\mathcal{E} = \{q : q_{\min} \leq |q| \leq q_{\max}\}$ with $0 < q_{\min} < q_{\max} < \infty$. Suppose $\alpha_0 \in I$ and $q_0 \in \text{int } \mathcal{E}$. If the two rate responses admit independent complex morphology disks with radii $\rho_1, \rho_2 \geq 0$ and $\rho_1 + \rho_2 > 0$, then the feasible order set contains a nondegenerate relative neighborhood of α_0 (one-sided at an endpoint).*

Proof idea. Write $c_r(\beta) = Z_r[\lambda_0 + (i\nu_r)^\beta]$ and let $F = c_1 - c_2$. Since $F(\alpha_0) = 0$, a local Lipschitz bound gives $|F(\beta)| \leq \rho_1 + \rho_2$ for all sufficiently close β . The weighted centre

$$q_w(\beta) = \frac{\rho_2 c_1(\beta) + \rho_1 c_2(\beta)}{\rho_1 + \rho_2}$$

lies in both disks and remains in the morphology annulus by continuity. Every such β therefore has a common witness. $\square$

Finally, finite frequency data do not identify a fractional mechanism. For L distinct conjugate-closed nodes $s_j = i\omega_j$ and consistent targets z_j , any $a > 0$ and integer $N \geq L$ permit polynomial interpolation of $z_j(s_j + a)^N$. The result is a real-coefficient, stable, strictly proper rational function $P(s)/(s + a)^N$ matching all nodes because $\deg P \leq L - 1 < N$. Even a certified singleton inside (2) is therefore an effective surrogate order, not proof of fractional cardiac physiology.

5 Finite-record set certificate

5.1 Exact joint disk inversion

Let the shared morphology coefficient for harmonic m belong to a compact annulus $\mathcal{E}_m = \{q : q_{\min,m} \leq |q| \leq q_{\max,m}\}$ with $q_{\min,m} > 0$. Write $\mathbb{D}(c, r) = \{z \in \mathbb{C} : |z - c| \leq r\}$ for a closed complex disk. For candidate (β, ℓ) define

$$d_{r,m}(\beta, \ell) = \ell + (i\nu_{r,m})^\beta, \quad \mathcal{Q}_{r,m} = \mathbb{D}(d_{r,m} \tilde{Z}_{r,m}, |d_{r,m}| \epsilon_{r,m}).$$

Theorem 5.1 (Forward-disk equivalence). *For fixed (β, ℓ) , one $q_m \in \mathcal{E}_m$ satisfies all rate-wise response disks if and only if*

$$\left(\bigcap_r \mathcal{Q}_{r,m}(\beta, \ell) \right) \cap \mathcal{E}_m \neq \emptyset.$$

Consequently the zero-drift order set is

$$\mathcal{A}_{\text{joint}} = \text{proj}_{\beta} \{(\beta, \ell) \in I \times \Lambda : (\bigcap_r \mathcal{Q}_{r,m}) \cap \mathcal{E}_m \neq \emptyset \ \forall m \in \mathcal{M}\}. \quad (9)$$

Proof. Because $d_{r,m} \neq 0$, the response-disk inequality is equivalent to $|d_{r,m} \tilde{Z}_{r,m} - q_m| \leq |d_{r,m}| \epsilon_{r,m}$. Sharing one q_m across rates gives the intersection; projection gives the order set. $\square$

Independent complex morphology restitution of radius $\rho_{r,m}^q$ is exact after adding $\rho_{r,m}^q$ to the corresponding q -disk radius. Correlated, real-only, or cross-harmonic restitution must retain its shared variables; disk inflation is then only an outer relaxation.

5.2 Conditional data-to-disk theorem

Suppose the finite-window signal has the decomposition

$$\mathbf{y} = \Psi \text{diag}(\hat{H}_k) \mathbf{Z} + \mathbf{r}, \quad \|\mathbf{r}\|_W \leq \bar{r}, \quad G \succ 0.$$

The WLS row norm is $a_m = \sqrt{(G^{-1})_{mm}}$, so Cauchy–Schwarz gives

$$|\tilde{Z}_m - Z_m| \leq \frac{a_m}{|\hat{H}_m|} \bar{r}. \quad (10)$$

The central issue is therefore not fitting $\bar{r}$, but certifying it without using the same target-window residual circularly.

For the delayed-causal protocol used here, a sufficient disk radius is

$$\begin{aligned} \epsilon_{r,m} = & \frac{a_m}{|\hat{H}_m|} (\bar{r}_{\text{hist}} + \bar{r}_{\text{dwell}} + \bar{r}_{\text{HRV}} + \bar{r}_{\text{phase}} + \bar{r}_{\text{sample}} + \bar{r}_{\text{noise}} + \bar{r}_{\text{model}}) \\ & + \epsilon_{r,m}^{\text{omitted}} + \epsilon_{r,m}^{\text{delay}} + \epsilon_{r,m}^{\text{quad}} + \epsilon_{r,m}^{\text{filt-init}} + \rho_{r,m}^{\text{num}}. \end{aligned} \quad (11)$$

History and dwell terms are obtained from the Mittag–Leffler variation-of-constants kernel; the HRV term propagates a phase-warp input bound through the same kernel. Sampling requires timestamp, ADC, interpolation, and anti-alias bounds. Front-end terms require complex response and relative-delay calibration. Discrete WLS noise needs a discrete weighted-norm contract; a continuous L_2 noise bound alone cannot control point samples without a sampling/reconstruction theorem. Trapezoidal continuous lock-in uses one internally generated composite-trapezoid weight vector and applies its quadrature remainder only to the declared nominal retained reconstruction. The term $\rho_{r,m}^{\text{num}}$ is an outward bound between the exact frozen sample-functional coefficient and its stored binary64 centre; it is not an acquisition-noise allowance.

Theorem 5.2 (Conditional finite-record coverage). *Under Assumption 3.1, $G \succ 0$, a calibrated nonzero $\hat{H}_m$, and simultaneously valid, independently sourced bounds for every term in (11), the disk (4) contains the true complex response. The sum is a valid outer disk, although it need not be minimal when the errors are correlated.*

Proof idea. Split the fractional convolution at the record start and rate switch to bound history and dwell. Propagate the phase perturbation as an input through the positive Mittag–Leffler kernel. Apply the declared extraction functional to the remaining sampling, front-end, noise, initialization, and discrepancy terms. The triangle inequality followed by (10) yields (11); the separately verified exact-payload enclosure adds the final stored-centre term $\rho_{r,m}^{\text{num}}$ without reusing a signal residual. $\square$

Complete monotonicity of $E_\alpha(-x)$ for $0 < \alpha \leq 1$ [24, 25] supplies computable uniform history/dwell envelopes. Finite-memory correction and truncation are related, but already established, control topics [18, 19]; the relevant contribution here is their placement in a non-overlapping response-disk budget.

5.3 Machine-verifiable composition

The implementation separates mathematical coverage from evidence integrity. The WLS execution record binds the protocol, source-record manifest, estimator manifest, window geometry, selected payload, times, weights, basis, G , right-hand side, solved coefficients, front-end response, and centre hashes. For the current frozen uniform-weight path, $w_n = 1/n_s$, an exact integer payload has a separate Arb/ACB certificate [29]. It reconstructs the phase and complete two-sided basis, accumulates containing G and right-hand-side balls directly from exact integer/dyadic inputs, performs a verified joint solve, and publishes $\rho_{r,m}^{\text{num}}$ as an outward dyadic upper bound. Arbitrary binary64 execution weights do not gain this exact-functional claim without a separate exact-weight certificate. A determinant or solve that cannot exclude singularity returns `NOT_CERTIFIABLE`.

The R2 component certificate binds the named computed subset—Gram geometry, history, dwell, sampling, optional quadrature, and relative-delay terms. The sealed protocol/result binds every applicable primitive and the total radius, including externally supplied HRV, phase-anchor, noise, model-discrepancy, harmonic-tail, and filter-initialization terms. A fast exact-rational Gershgorin lower bound is supplemented by a ball-arithmetic Hermitian LDL^* proof of $G - \mu I \succ 0$ when Gershgorin is inconclusive.

Here R2 denotes the data-to-response-disk coverage gate. R1 is the downstream joint analytic-selector witness on the parameter box. `PRESERVED_ROBUST` means that the old R1 inner witness retains strictly positive certified margin under the new disks. `PRESERVED_CLOSED` means only nonnegative retained margin. Neither relation is an outer feasibility proof or general model-acceptance decision.

The sealed downstream bundle replays the protocol, one component link per canonical rate-harmonic target, the previous analytic-selector tube, and a margin-transfer proof. It reconstructs the response-disk family and checks centre movement, radius floors, unit/split consistency, priors, rate geometry, and exact result-set equality. It has no rejection output: failure of this inner-witness composition is not an outer proof that the model is impossible.

5.4 Worst-case ambiguity and design

Let $\mathfrak{D}$ be a compact design set, Θ a compact nuisance set, and $f_{\mathcal{D}}(\alpha, \theta) \in \mathbb{C}^n$ a jointly continuous nominal map in $(\mathcal{D}, \alpha, \theta)$, with the same finite output dimension n for every design. Let the continuous product-disk radii satisfy $\epsilon_j(\mathcal{D}) \geq \epsilon_{\min} > 0$, and define

$$\mathcal{Y}_{\mathcal{D}}(\alpha) = \{f_{\mathcal{D}}(\alpha, \theta) + e : \theta \in \Theta, |e_j| \leq \epsilon_j(\mathcal{D})\}, \quad \mathcal{A}_{\mathcal{D}}(y) = \{\alpha : y \in \mathcal{Y}_{\mathcal{D}}(\alpha)\}.$$

Write $\mathcal{O}_{\mathcal{D}} = \cup_{\alpha \in I} \mathcal{Y}_{\mathcal{D}}(\alpha)$ and let $\mathfrak{R}_{\mathcal{D}}$ be the class of deterministic set estimators that contain the true order for every admissible (α, θ, e) .

Theorem 5.3 (Minimax ambiguity and exact disk threshold). *The smallest achievable worst-case diameter is*

$$\begin{aligned} \inf_{S \in \mathfrak{R}_{\mathcal{D}}} \sup_{y \in \mathcal{O}_{\mathcal{D}}} \text{diam } S(y) &= \sup_{y \in \mathcal{O}_{\mathcal{D}}} \text{diam } \mathcal{A}_{\mathcal{D}}(y) \\ &= \Delta(\mathcal{D}), \\ \Delta(\mathcal{D}) &= \max_{\substack{\alpha, \beta \in I \\ \mathcal{Y}_{\mathcal{D}}(\alpha) \cap \mathcal{Y}_{\mathcal{D}}(\beta) \neq \emptyset}} |\alpha - \beta|. \end{aligned} \tag{12}$$

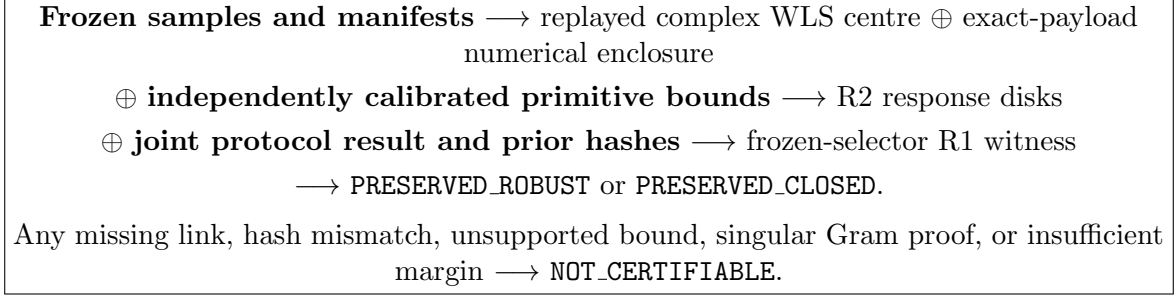


Figure 1: Certificate composition and fail-closed semantics. The current real-data pilot completes WLS execution replay but not the independent R2 primitive calibration; it therefore stops before a real-data response-disk pass.

For $0 < \delta \leq \text{diam } I$, let

$$\mu_{\mathcal{D}}(\delta) = \min_{\substack{\alpha, \beta \in I \\ |\alpha - \beta| \geq \delta}} \min_{\theta, \theta' \in \Theta} \max_j \frac{|f_{\mathcal{D},j}(\alpha, \theta) - f_{\mathcal{D},j}(\beta, \theta')|}{\epsilon_j(\mathcal{D})}.$$

Then $\Delta(\mathcal{D}) < \delta$ if and only if $\mu_{\mathcal{D}}(\delta) > 2$. Moreover, $\mathcal{D}_{\delta}^* \in \arg \max_{\mathcal{D} \in \mathfrak{D}} \mu_{\mathcal{D}}(\delta)$ exists. If this optimum is at most two, no uniformly reliable deterministic set estimator under any admissible design can guarantee diameter below δ .

Proof idea. Every order in $\mathcal{A}_{\mathcal{D}}(y)$ can be the truth for an admissible realization producing y , so any reliable estimator must contain the whole consistency set. Two product disks intersect exactly when every normalized centre separation is at most two; their midpoint proves sufficiency. Compact continuity gives existence of the optimum. $\square$

For log-rate spacing $h \downarrow 0$, the local worst-direction sensitivity of the physical two-rate inverse is $\Theta(h^{-1})$. A particular three-rate difference-quotient diagnostic is $\Theta(h^{-2})$; this is a condition number of that diagnostic, not a lower bound for all three-rate estimators. The design theorem therefore optimizes rate separation and certified R2 radii together rather than using a fixed beats-per-minute rule.

6 Reproducible numerical evidence

6.1 Synthetic and adversarial verification

Table 1 summarizes the deterministic evidence bundle. All random experiments use NumPy PCG64 with seed 20260711, and stored CSV outputs are byte-reproducible apart from explicitly excluded elapsed-time fields. These experiments check implementation consistency and bound coverage on their declared domains; they do not replace a proof or calibrate real ECG errors.

The production exact-sample case then closes the implemented composition on eight signed integer samples at indices 80–87 with uniform weights. WLS execution replayed **MATCH**; the independent full exact-functional certificate replayed **MATCH/CERTIFIED_INTERVAL**; the R2 component was **CERTIFIED_BOUND**; the frozen protocol was **PASS_DETERMINISTIC**; and the bundle replayed **PRESERVED_ROBUST**. Its known-model response-to-centre error was 6.51×10^{-11} , while the published $1.00000004 \times 10^{-9}$ disk contained the truth and had radius-to-response-magnitude ratio 3.80×10^{-9} . Changing either one digital sample or the interval radius without a valid seal produced **NOT_CERTIFIABLE**. The ordinary binary64 WLS record alone deliberately retains **NOT_CERTIFIABLE**; only the separately replayed exact uniform-weight interval closes its phase, G , right-hand-side, solve, and division rounding. This positive chain is synthetic and does not supply any real-data primitive.

Table 1: Synthetic and replay evidence. “Pass” is scoped to the test in the second column and is not a real-ECG order certificate.

Layer	Check	Cases	Result
Two-rate theorem	Global inverse and common-gain invariance	1,260	all pass; max. $ \Delta\alpha = 4.07 \times 10^{-13}$
Three-rate check	Reference-pair consistency / valid-pair tamper	1,260/1,260	all consistent / all inconsistent
Exact sample chain	WLS/interval/R2/protocol/bundle; two seal attacks	1+2	preserved robust; both attacks not certifiable
Legacy 3-rate	Analytic triplet recovery diagnostic	1,200	1,200 pass
Full memory	Pointwise history/dwell and demodulation bounds	75+15	all pass
Switch stress	Constant positive/negative forcing switches	225	all pass; max utilization 0.999832
Legacy G0 gate	Valid controls / invalid protocols / equivalent nuisances	60/180/10	60 pass; 190 not certifiable
Legacy triplet outer	Bounded random complex perturbations	49,152	0 bound violations
Legacy triplet boundary	Circle-directed adversarial perturbations	256	100% order coverage

The production two-rate sweep used a fixed Cartesian grid over 21 orders from 0.05 to 1, five damping decades from 10^{-2} to 10^2 , four rate ratios from 1.25 to 6, and three fixed (base-frequency, nonzero-complex-morphology) pairs; four common gains were cycled deterministically. It supplied the global solver only the fixed bracket $[10^{-6}, 1]$, never a truth-centred initialization. All 1,260 inversions and common-complex-gain checks passed; the maximum damping and morphology errors were 8.86×10^{-10} and 2.19×10^{-11} . All positive geometric three-rate triples were consistent. Replacing only the third response by a separately valid pair with 5% changed damping produced 1,260/1,260 explicit INCONSISTENT results, rather than relying on an out-of-image failure. This is a floating-point implementation audit of the proved identities, not a proof and not noisy-data evidence.

The legacy three-rate diagnostic’s maximum absolute recovery errors were 3.47×10^{-13} for α , 2.78×10^{-12} for λ , and 4.58×10^{-12} for q . In bounded propagation the maximum observed error-to-bound utilizations were 0.999968 for inversion $Z \mapsto W$, 0.960317 for the next quotient layer, and 0.884876 for the order layer. The near-one values show that the tests include nearly saturating directions rather than only benign central perturbations. These triplet calculations are a conservative fixed-subset outer propagation diagnostic, not the exact joint R1 feasible-set projection and not the two-rate quotient inverse of Theorem 4.2.

The same legacy outer diagnostic deteriorated rapidly even though its declared coverage remained correct. At relative complex-response radii 0.01%, 0.1%, 0.5%, and 1%, its median conservative widths were respectively 0.00659, 0.06436, 0.25198, and 0.44545. The fraction meeting a fixed width-0.02 target fell from 100% to 25%, 0%, and 0%. Structural identifiability therefore does not by itself guarantee a narrow certified outer set. These widths diagnose the conservatism and practical resolution of that legacy propagation chain, not the exact ambiguity of the joint feasible set.

The certificate test suite also modifies or removes evidence fields. Missing rate-harmonic links, component/record/estimator hash mismatches, altered units or split provenance, inconsistent priors, understated radii, overstated Gram bounds, duplicate JSON keys, centre displacement beyond the frozen margin, and aliased singular Gram matrices are all downgraded to NOT_CERTIFIABLE. A positive-definite but non-diagonally-dominant Gram example exercises

the validated LDL^* fallback.

6.2 Design-locked four-record Fantasia pilot

Fantasia contains long ECG and respiration recordings from young and older participants and was originally used to study age-related interbeat dynamics [26]. We used Fantasia Database v1.0.0 [27, 28], DOI 10.13026/C2RG61, under its Open Data Commons Attribution License v1.0. Four pre-specified records came from the already local, read-only collection: f1y01 and f1o01 for engineering calibration, and f2y01 and f2o01 for locked engineering validation. This is a pipeline pilot, not a clinical cohort comparison.

Within each subject, the interval before 1,800 s was used for identification and the interval after 2,100 s for validation, leaving a 300 s guard interval. Windows span 16 adjacent RR intervals. The fixed two-sided basis is $\mathcal{K}_{\text{est}} = \{-3, -2, -1, 0, 1, 2, 3\}$, $\tau_\star = 1$ s, and $\hat{H}_m = 1$. Windows longer than 4,096 samples use a design-locked, value-independent index selection; 233/1,859 windows used that path. Since no anti-alias bound is available for this selection, it is explicitly a blocker for R2 certification.

The derived package contains no raw waveform. It stores source hashes, window manifests, compressed coefficients, model comparisons, and compressed WLS execution records; the latter retain selected beat-sample indices derived from the licensed annotations, so the dataset attribution remains applicable. Rereading the external records reproduced 1,859/1,859 execution hashes. Relative to the recorded binary64 G and right-hand side, ball arithmetic enclosed the joint solve and identity-front-end division in 1,859/1,859 windows; the maximum outward absolute radius was 3.62×10^{-15} . That radius does not cover binary64 accumulation of G and the right-hand side, nor any acquisition/model primitive.

Four design-locked predictors were compared on the temporal holdout: fractional-common (α, λ) with harmonic-specific q_m , the nested $\alpha = 1$ model, a rate-independent complex Fourier mean, and a rate-shuffle negative control. Table 2 reports the validation RMSE.

For windows w and harmonics $m = 1, 2, 3$, we fit the complex coefficients by minimizing

$$\text{SSE} = \sum_{w,m} |\tilde{Z}_{w,m} - \hat{Z}_{w,m}|^2, \quad \text{RMSE} = \sqrt{\text{SSE}/n_{\mathbb{C}}},$$

where $n_{\mathbb{C}}$ is the number of complex coefficients. At fixed (α, λ) each complex q_m is profiled by its analytic least-squares solution. The search evaluates a 17×25 grid on $\alpha \in [0.05, 1]$ and $\log \lambda \in [\log(10^{-3}), \log(10^3)]$, then refines the six best starts with bounded L-BFGS-B. These are returned local-search solutions, not certified global optima. The seeded shuffle control permutes only the identification rate-window pairing, records the permutation hash, and predicts the untouched temporal holdout at its actual rates.

The material-improvement threshold was fixed at 0.1% relative RMSE. All four returned free-order fits terminated at $\alpha = 1$, and the fractional/nested RMSE ratio was numerically one in 4/4 records. Thus the fractional model had zero material wins over its $\alpha = 1$ submodel. On locked f2y01 it improved on the constant and shuffle controls by about 1.30% and 1.23%, whereas on locked f2o01 it was worse by about 2.71% and 2.72%. The directions disagree. The identification/validation heart-rate ranges (Hz) were respectively 1.105–1.421/1.048–1.581 (f1y01), 0.941–1.100/0.911–1.219 (f1o01), 1.085–1.292/1.030–1.475 (f2y01), and 1.038–1.351/0.927–1.152 (f2o01). The returned fractional-model λ values were 32.812, 0.001, 2.0045, and 0.001; the f1o01 and f2o01 solutions hit the predeclared lower bound. These boundary hits and the uncontrolled, substantially overlapping rate ranges further limit interpretation. Accordingly,

distinct fractional-order evidence = NOT_SUPPORTED,

and confirmatory rate-response evidence is also NOT_SUPPORTED.

Table 2: Fantasia temporal-holdout RMSE (mV). f1 records are engineering calibration records; f2 records are locked engineering-validation records.

Record	Fractional	$\alpha = 1$	Constant	Rate shuffle
f1y01	0.056631	0.056631	0.056638	0.056688
f1o01	0.020851	0.020851	0.020974	0.020920
f2y01	0.019044	0.019044	0.019296	0.019282
f2o01	0.009714	0.009714	0.009458	0.009457

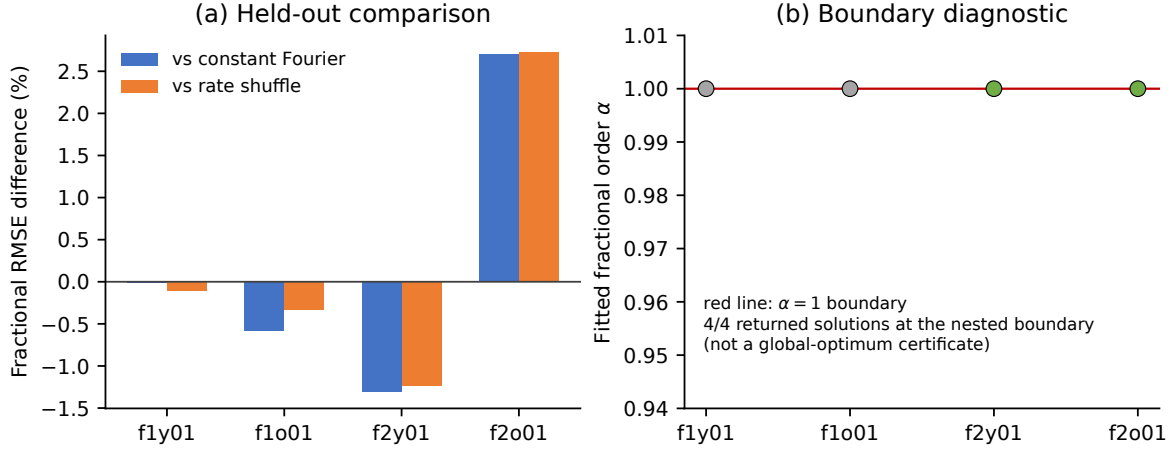


Figure 2: Design-locked Fantasia engineering pilot. Every fitted free order lands at the admissible boundary $\alpha = 1$. A negative RMSE percentage difference favors the fractional fit; the two locked records disagree in direction against the constant and rate-shuffle controls. The figure is descriptive and does not represent a certified R2 inference.

All four real records remain **NOT_CERTIFIABLE**. Missing primitives include annotation-timing and whole-window phase-warp error, measurement-noise and model-discrepancy envelopes, finite-dwell and prehistory bounds, sampling/quadrature and subsampling anti-alias bounds, omitted-harmonic tail/leakage, ADC-gain and analog-front-end calibration uncertainty, and a complete interval enclosure of G and the right-hand-side accumulation. Empirical validation residuals are not used to fill these bounds.

7 Discussion

7.1 What the theory establishes

The global quotient theorem provides a clean answer to the structural question posed in the introduction. Positive real damping adds a phase constraint that makes two complete complex rate responses sufficient despite the unknown complex morphology. This is stronger than counting complex equations, because it identifies the complete nuisance quotient and its physical image. The multi-rate result then separates estimation from model checking: two rates are exactly determined; a third rate creates redundancy.

The result is useful only when the common-gauge protocol is credible. Theorems 4.6 and Proposition 4.5 show why morphology handling cannot be an informal preprocessing choice. Allowing arbitrary complex changes removes all order information, and even a small positive independent disk prevents a singleton conclusion in the stated two-rate problem. These are not merely numerical conditioning warnings; they are identifiability limits.

The finite-record layer gives a second separation. WLS geometry converts a valid residual envelope to a coefficient disk, but cannot create that envelope. Hardware, annotations, front-end

response, memory tails, and model discrepancy need independent evidence. Fail-closed replay protects this boundary against accidentally reusing a fitted residual as both estimate and coverage proof.

7.2 Interpretation of the negative pilot

The Fantasia result does not refute every fractional ECG model. It says that, under this frozen scalar harmonic family and four-record engineering design, the additional order freedom did not separate from the $\alpha = 1$ boundary. Nor did the two locked records show a consistent advantage over simpler controls. This is precisely the kind of result that a nested baseline and negative control should reveal before a larger study.

The useful positive outcome is narrower: fixed-phase segmentation, two-sided joint complex WLS, source/payload hashing, deterministic replay, temporal guarding, and model-comparison code all operated on long external ECG records. Those engineering checks justify a future controlled multi-rate acquisition; they do not justify a physiological-order claim.

7.3 Limitations and next experiments

First, the scalar steady harmonic surrogate omits multichannel propagation, beat morphology dynamics, autonomic covariates, and spatial cardiac states. Its order should not be interpreted as tissue-level memory. Second, the finite-frequency rational-interpolation result prevents mechanism attribution from harmonic agreement alone. Held-out transients and integer-order rational baselines are necessary.

Third, the current real-data evidence closes WLS execution replay but not R2 coverage. The missing primitive bounds listed above must be obtained through instrument calibration, controlled annotation studies, interval reconstruction, and an independently specified model-discrepancy contract. Fourth, the four records are too few for population inference and only two were locked engineering-validation records. No age-group conclusion is drawn. Fifth, the observed heart-rate range was not optimized using Theorem 5.3; uncontrolled resting data can be structurally valid yet practically uninformative.

A decisive follow-up should therefore use controlled, sufficiently separated rate plateaus; one fixed acquisition chain; explicit dwell and prehistory control; calibration data disjoint from subject-specific identification and future validation; and a predeclared R2 budget. The experiment should proceed only if the optimized separation modulus exceeds two at the target order resolution. A larger ECG cohort or learned centre estimator is secondary: neither can replace missing coverage primitives.

8 Conclusion

We have presented a control-theoretic identifiability and certification framework for a fractional surface-ECG harmonic surrogate. The core theorem shows that two exact complex rate responses uniquely recover order and damping modulo a common complex morphology/lead gauge, while one response is insufficient. Joint response disks, independently calibrated finite-record bounds, validated Gram margins, and fail-closed replay extend the structural result toward auditable data analysis. The accompanying no-go and minimax results define when the problem cannot deliver a narrow order set and how a future rate-window experiment should be designed.

The reported real-data pilot is intentionally retained as a negative result: all four fitted orders equal one, the nested integer-order model is not improved, and the necessary real-data error bounds are incomplete. The current conclusions are therefore `NOT_SUPPORTED` for distinct fractional order and `NOT_CERTIFIABLE` for real-ECG set inference. Preserving those conclusions is essential to the credibility of any later, larger validation.

Reproducibility and evidence availability

The repository contains the theory-core supplement, protocol template, source-code certificate issuers/replayers, deterministic test scripts, the sealed exact-sample and two-rate summaries, and the derived Fantasia evidence bundle. Raw ECG records remain outside the repository; all 12 selected files match the official Fantasia v1.0.0 SHA-256 manifest. The Fantasia bundle contains only manifests, derived coefficients, execution records (including derived beat indices), comparisons, and checksums under the source-dataset attribution boundary. Production-source hashes, package/runtime identifiers, and reproduction commands are recorded with the evidence objects. The main manuscript does not treat generated point estimates as certified disks.

References

- [1] P. E. McSharry, G. D. Clifford, L. Tarassenko, and L. A. Smith, “A dynamical model for generating synthetic electrocardiogram signals,” *IEEE Transactions on Biomedical Engineering*, vol. 50, no. 3, pp. 289–294, 2003. doi:10.1109/TBME.2003.808805.
- [2] R. Sameni, M. B. Shamsollahi, C. Jutten, and G. D. Clifford, “A nonlinear Bayesian filtering framework for ECG denoising,” *IEEE Transactions on Biomedical Engineering*, vol. 54, no. 12, pp. 2172–2185, 2007. doi:10.1109/TBME.2007.897817.
- [3] D. Sierociuk and A. Dzieliński, “Fractional Kalman filter algorithm for the states, parameters and order of fractional system estimation,” *International Journal of Applied Mathematics and Computer Science*, vol. 16, no. 1, pp. 129–140, 2006.
- [4] D. Sierociuk and M. Macias, “Triple estimation of fractional variable order, parameters, and state variables based on the unscented fractional order Kalman filter,” *Sensors*, vol. 21, no. 23, 8159, 2021. doi:10.3390/s21238159.
- [5] W. Yu, Y. Zhang, R. Chen, W. Zhu, C. Wen, and Y. Chen, “A fractional high-order extended Kalman filter for fractional-order nonlinear systems,” *ISA Transactions*, vol. 171, pp. 152–162, 2026. doi:10.1016/j.isatra.2026.02.020.
- [6] N. Nazarian, M. Haeri, and M. S. Tavazoei, “Identifiability of fractional order systems using input output frequency contents,” *ISA Transactions*, vol. 49, pp. 207–214, 2010. doi:10.1016/j.isatra.2009.11.007.
- [7] M. S. Tavazoei and M. Haeri, “A proof for non existence of periodic solutions in time invariant fractional order systems,” *Automatica*, vol. 45, no. 8, pp. 1886–1890, 2009. doi:10.1016/j.automatica.2009.04.001.
- [8] J.-S. Duan, Y.-J. Lan, and M. Li, “A comparative study of responses of fractional oscillator to sinusoidal excitation in the Weyl and Caputo senses,” *Fractal and Fractional*, vol. 6, no. 12, art. 692, 2022. doi:10.3390/fractalfract6120692.
- [9] R. Malti, T. Raïssi, M. Thomassin, and F. Khemane, “Set membership parameter estimation of fractional models based on bounded frequency domain data,” *Communications in Nonlinear Science and Numerical Simulation*, vol. 15, pp. 927–938, 2010. doi:10.1016/j.cnsns.2009.05.005.
- [10] F. Khemane, R. Malti, T. Raïssi, and X. Moreau, “Robust estimation of fractional models in the frequency domain using set membership methods,” *Signal Processing*, vol. 92, pp. 1591–1601, 2012. doi:10.1016/j.sigpro.2011.12.008.
- [11] S. Abrashov, R. Malti, M. Moze, X. Moreau, F. Aioun, and F. Guillemard, “Simple and robust experiment design for system identification using fractional models,” *IEEE Transactions on Automatic Control*, vol. 62, pp. 2648–2658, 2017. doi:10.1109/TAC.2016.2614910.
- [12] M. Karimshoushtari and C. Novara, “Design of experiments for nonlinear system identification: A set membership approach,” *Automatica*, vol. 119, 109036, 2020. doi:10.1016/j.automatica.2020.109036.

- [13] S. Das and K. Maharatna, “Fractional dynamical model for the generation of ECG like signals from filtered coupled Van-der-Pol oscillators,” *Computer Methods and Programs in Biomedicine*, vol. 112, no. 3, pp. 490–507, 2013. doi:10.1016/j.cmpb.2013.08.012.
- [14] D. J. Templos-Hernandez et al., “A fractional-order approach to cardiac rhythm analysis,” *Chaos, Solitons & Fractals*, vol. 145, 110942, 2021. doi:10.1016/j.chaos.2021.110942.
- [15] A. Takha, M. L. Talbi, and P. Ravier, “Fractional calculus integration for improved ECG modeling: A McSharry model expansion,” *Medical Engineering & Physics*, vol. 132, 104237, 2024. doi:10.1016/j.medengphy.2024.104237.
- [16] K. Gillette et al., “A framework for the generation of digital twins of cardiac electrophysiology from clinical 12-leads ECGs,” *Medical Image Analysis*, vol. 71, 102080, 2021. doi:10.1016/j.media.2021.102080.
- [17] T. Grandits, K. Gillette, G. Plank, and S. Pezzuto, “Accurate and efficient cardiac digital twin from surface ECGs: Insights into identifiability of ventricular conduction system,” *Medical Image Analysis*, vol. 105, 103641, 2025. doi:10.1016/j.media.2025.103641.
- [18] A. Rauh and R. Malti, “Quantification of time-domain truncation errors for the reinitialization of fractional integrators,” *Acta Cybernetica*, vol. 26, pp. 105–128, 2023. doi:10.14232/actacyb.296010.
- [19] A. Rauh and M. Lahme, “Observer-based error correction in finite memory approaches for pseudo state estimation of fractional dynamic models,” *Annual Reviews in Control*, vol. 60, 101018, 2025. doi:10.1016/j.arcontrol.2025.101018.
- [20] A. Neumaier, “Rigorous sensitivity analysis for parameter-dependent systems of equations,” *Journal of Mathematical Analysis and Applications*, vol. 144, pp. 16–25, 1989. doi:10.1016/0022-247X(89)90357-0.
- [21] S. M. Rump, “Verification methods: Rigorous results using floating-point arithmetic,” *Acta Numerica*, vol. 19, pp. 287–449, 2010. doi:10.1017/S096249291000005X.
- [22] A. Goldsztejn and L. Jaulin, “Inner and outer approximations of existentially quantified equality constraints,” in *CP 2006*, LNCS 4204, pp. 198–212, 2006. doi:10.1007/11889205_16.
- [23] B. Ponleitner and H. Schichl, “Exclusion regions for parameter-dependent systems of equations,” *Journal of Global Optimization*, vol. 81, pp. 621–644, 2021. doi:10.1007/s10898-021-01082-3.
- [24] H. Pollard, “The completely monotonic character of the Mittag-Leffler function $E_\alpha(-x)$,” *Bulletin of the American Mathematical Society*, vol. 54, pp. 1115–1116, 1948. doi:10.1090/S0002-9904-1948-09132-7.
- [25] T. Simon, “Comparing Fréchet and positive stable laws,” *Electronic Journal of Probability*, vol. 19, paper 16, pp. 1–25, 2014. doi:10.1214/EJP.v19-3058.
- [26] N. Iyengar, C.-K. Peng, R. Morin, A. L. Goldberger, and L. A. Lipsitz, “Age-related alterations in the fractal scaling of cardiac interbeat interval dynamics,” *American Journal of Physiology—Regulatory, Integrative and Comparative Physiology*, vol. 271, no. 4, pp. R1078–R1084, 1996. doi:10.1152/ajpregu.1996.271.4.R1078.
- [27] C.-K. Peng and L. Lipsitz, “Fantasia Database,” PhysioNet, version 1.0.0, 2003. doi:10.13026/C2RG61.
- [28] A. L. Goldberger, L. A. N. Amaral, L. Glass, J. M. Hausdorff, P. Ch. Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C.-K. Peng, and H. E. Stanley, “PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals,” *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000. doi:10.1161/01.CIR.101.23.e215.
- [29] F. Johansson, “Arb: Efficient arbitrary-precision midpoint-radius interval arithmetic,” *IEEE Transactions on Computers*, vol. 66, no. 8, pp. 1281–1292, 2017. doi:10.1109/TC.2017.2690633.